Sintered Martian Basalt Regolith Structural Construction

1. Microwave Sintering Kinetics

Martian regolith simulant is heated to 1150 degrees Celsius using 2.45 GHz industrial magnetrons, fusing basalt grains without requiring external cement or water binders.

2. Compressive Strength Metrics

Sintered basalt structural blocks demonstrate compressive strength of 68 MPa, surpassing standard high-strength terrestrial concrete and resisting abrasive sand scouring.

3. Vaulted Arch Radiation Overburden

Robotic gantry printers deposit interlocking sintered arches over habitat inflatable pressure vessels, supporting a 3-meter regolith overburden to block galactic cosmic rays.